Spaces Protocol

Scalable & Permissionless Bitcoin Identities

Spaces empowers users with a sovereign Bitcoin identity, anchored securely on the Bitcoin blockchain. Imagine a social handle that is truly yours—censorship-resistant and accessible across various protocols like Nostr, wallet naming, peer-to-peer applications, and beyond.

Introduction

Uncancellable

Motivation

Use The Source

- https://explorer.spacesprotocol.org/
- https://docs.spacesprotocol.org/
- https://spacesprotocol.org/paper/
- https://github.com/spacesprotocol/sips

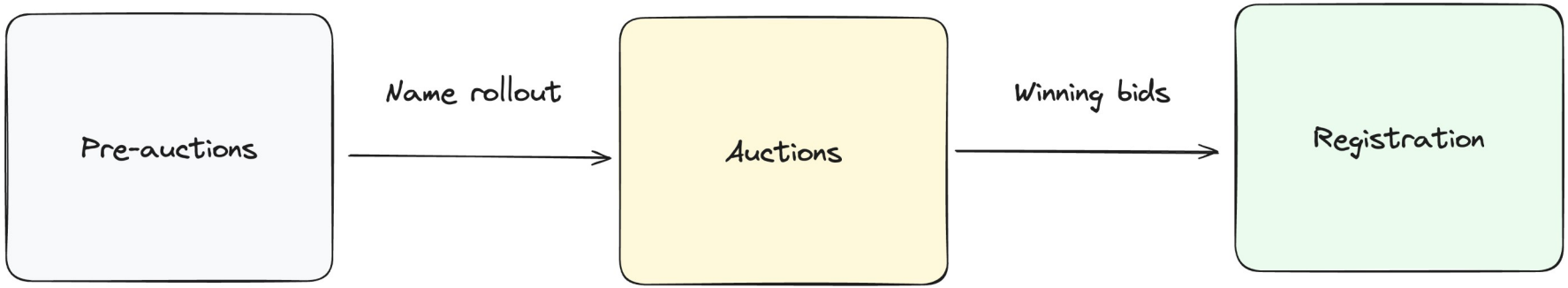
- https://explorer.spacesprotocol.org/space/bitcoin2026

Name Syntax

bob@bitcoin

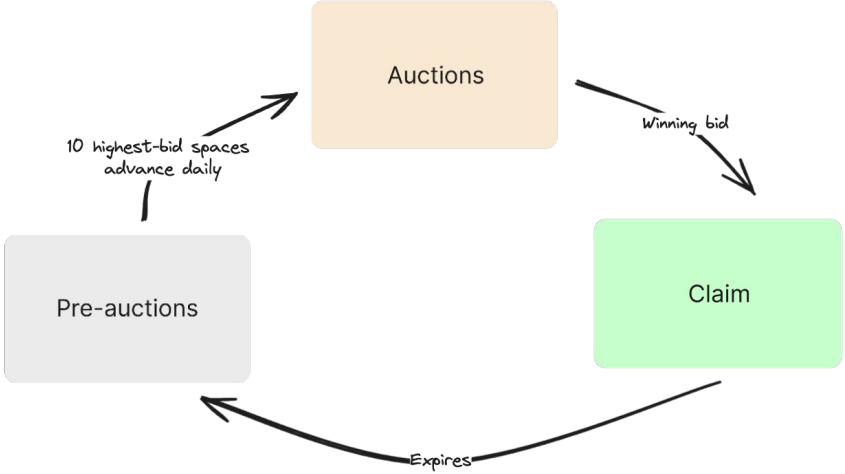
- Looks like an email address but isn’t
- Everything to the right of @ is on-chain
- No dots in the top-level name
- Everything to the left is zk proof+commit

Acquiring a Space



- All actions are real Bitcoin on-chain txs
- Open a name with initial bid (1000 sats)
- Only 10 new names(highest bid) daily
- Auctions last 10 days(extended)
- Highest bit uncontested gets to register

Top-Level Name Lifecycle



- Only 10 prevents request flooding
- 10+ day auction = fair open bidding
- Registered names expire in 365 days
- Any on-chain action resets the expiration

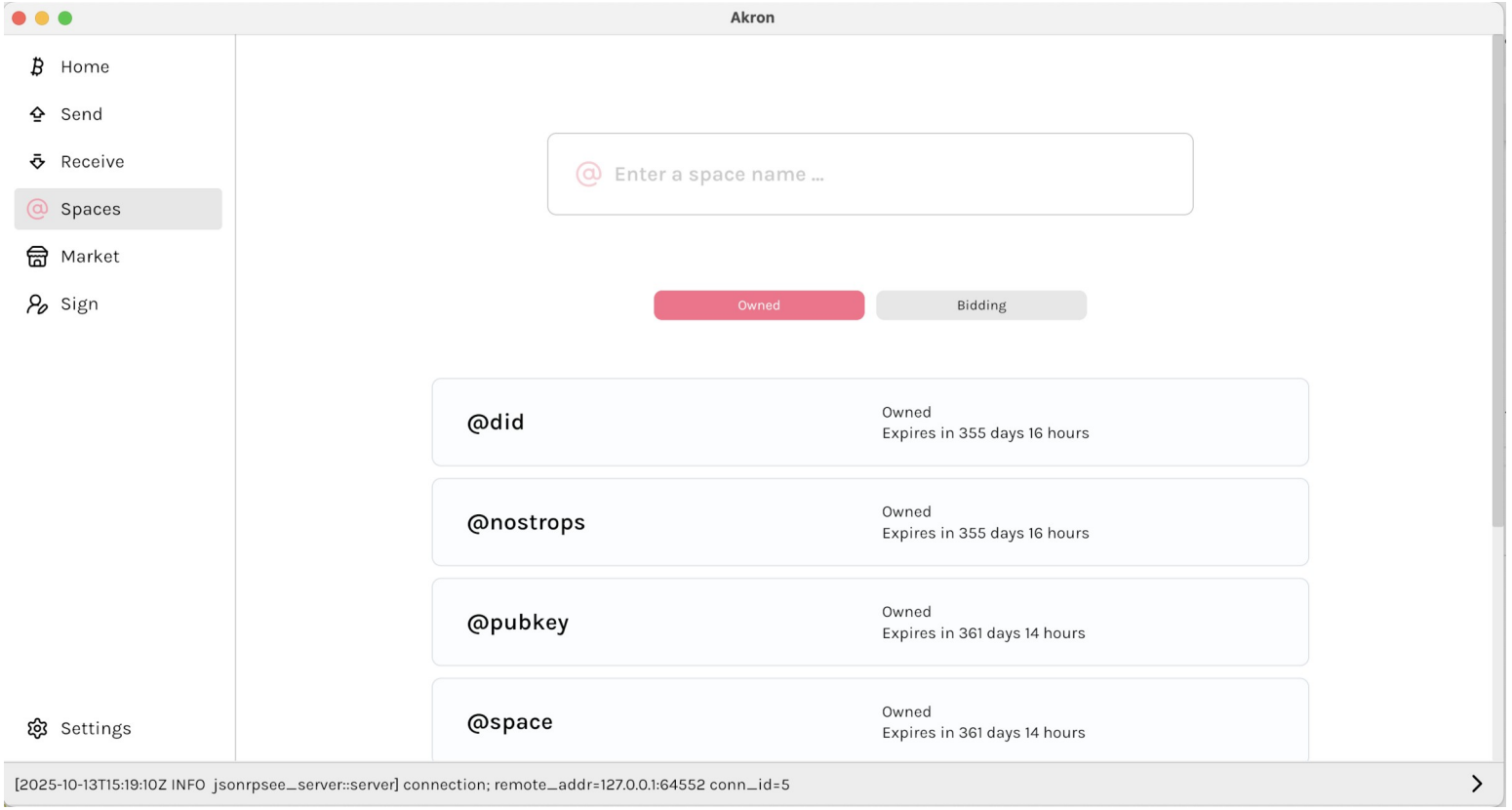
Spaced Indexer + CMD

- Crawls Mainnet looking for spaces txs
- Builds current state of top-level names
- Space-CLI command line operations
- Wallet “owns” the special UTXOs
- Build from source, Start9 & Umbrel

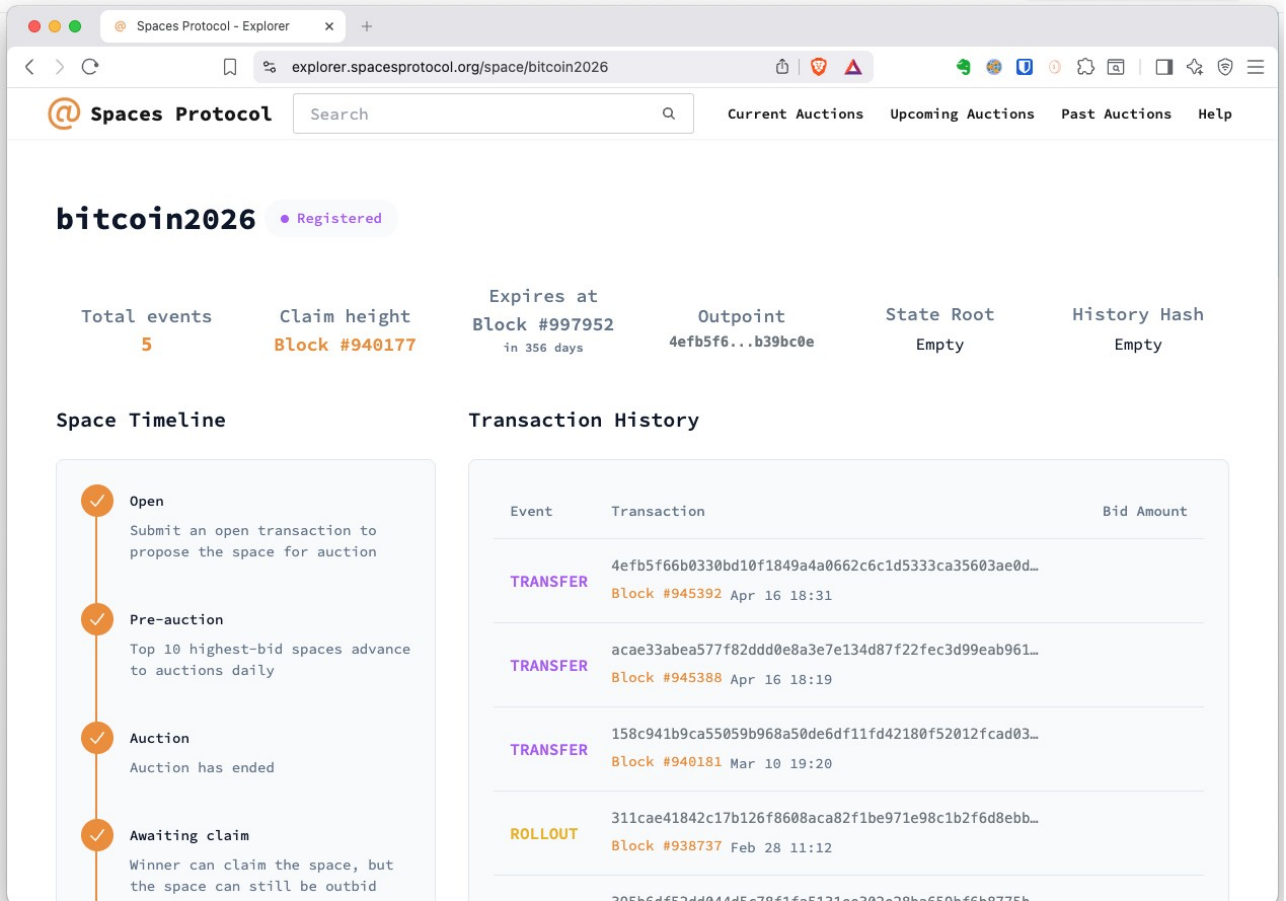
Akron Wallet

- Client Wallet Application
- Fund with Bitcoin
- Top-Level Name Operations
- Open, Bid, Register, Offer, Sell, Sign

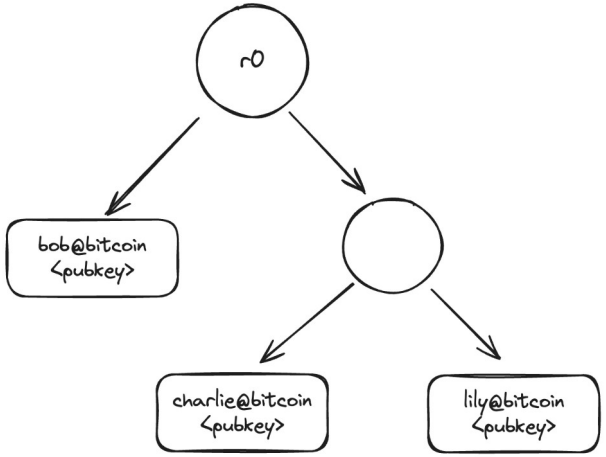
Akron Wallet



Spaces Explorer



Subspace Name Lifecycle



- Subspaces use a Merkle-ized structure
- Top-level owners “operate” the subspaces
- Subspaces can be given or sold

Subspace Names (Cont)

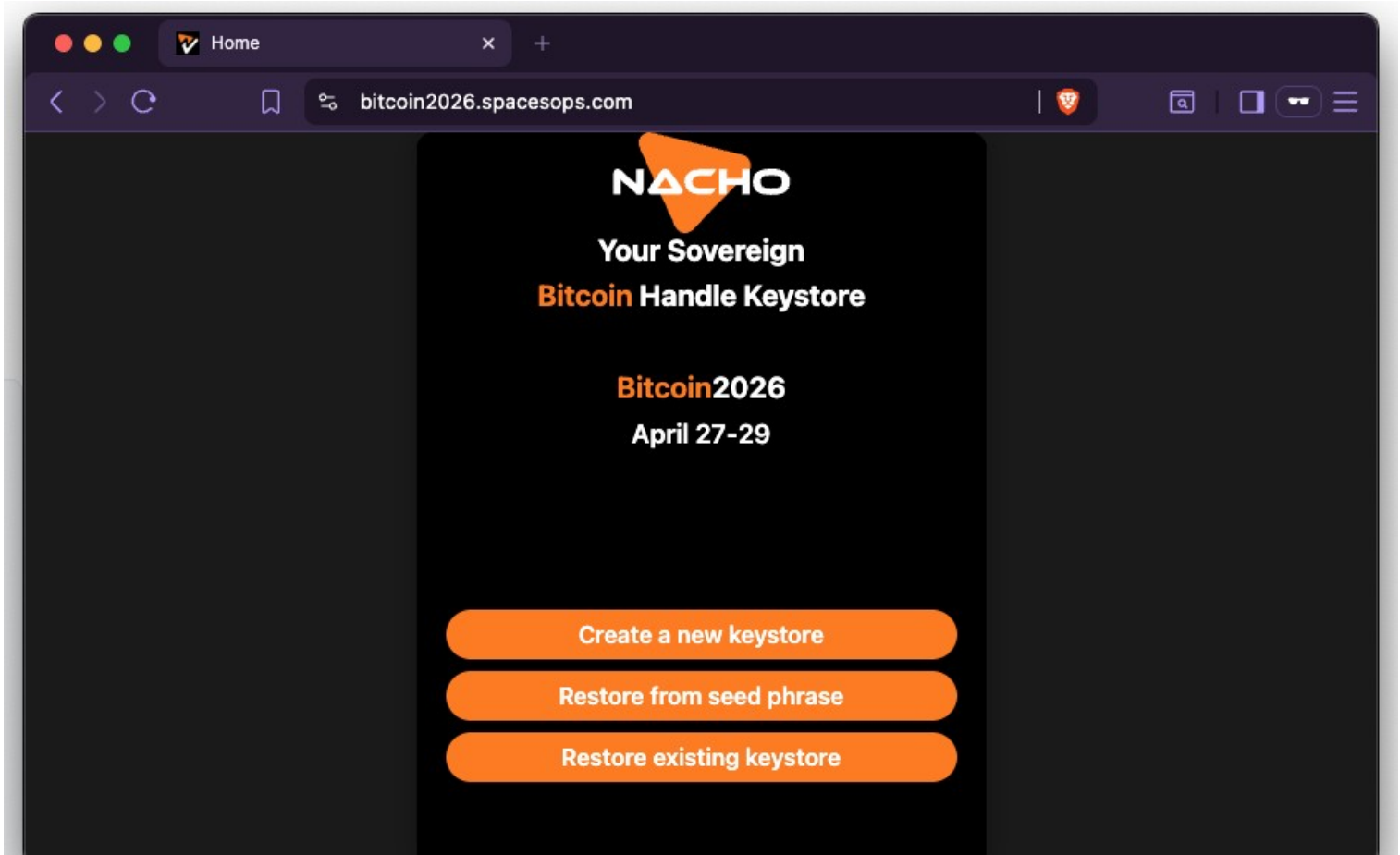
- Certificates and Meta-data resolved via a Cert-Relay node network
- Subnames can be validated off-chain
- Subnames NEVER expire
- Each subname has derived priv/pub keys

Nacho Wallet

- Web Based Wallet
- Android APK via Google Play Store
- iOS (or Android) via Expo Go App
- Manages “handle” requests

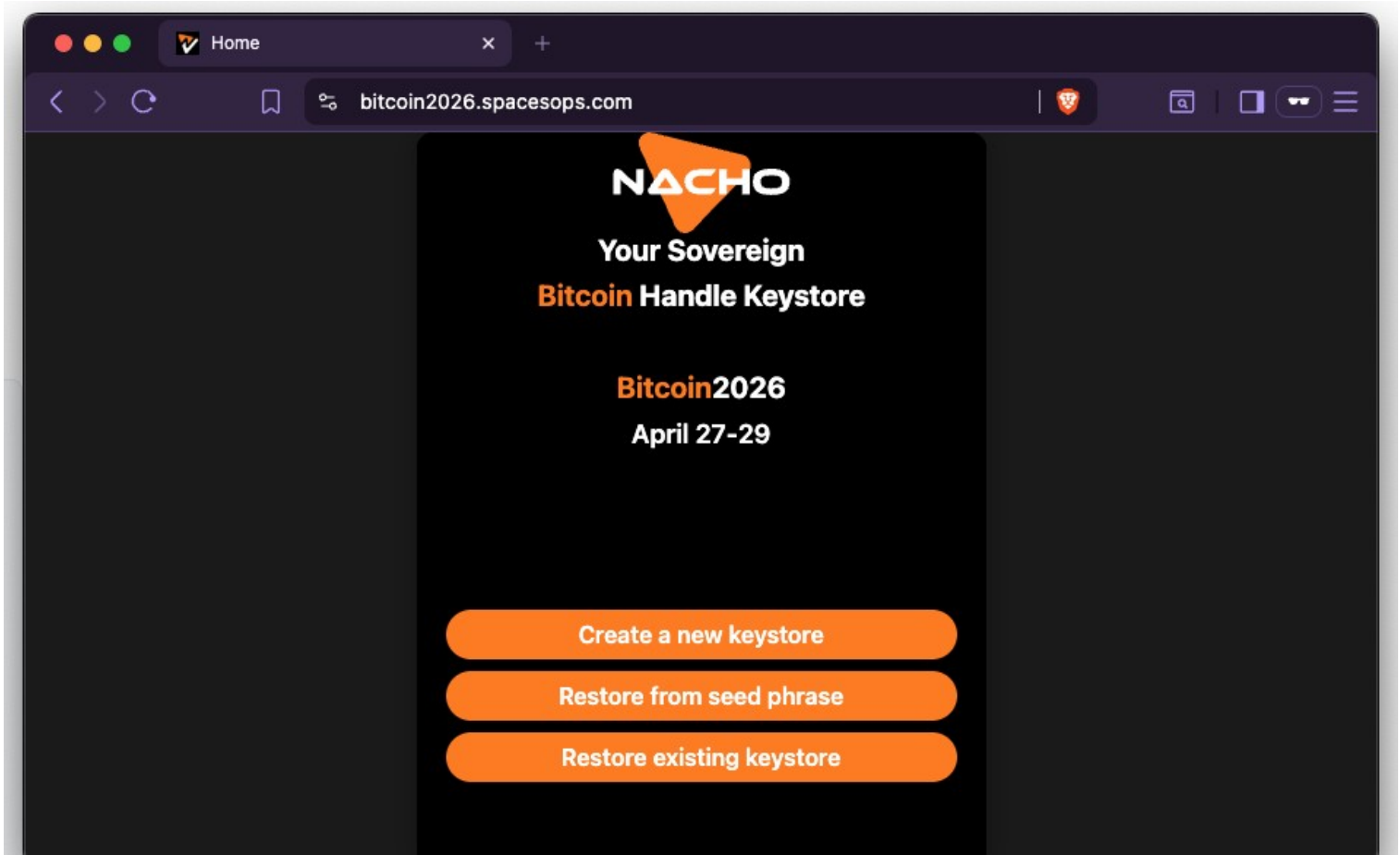
- Variant used for this workshop
- @Bitcoin2026 : closed after this week

Nacho Bitcoin2026 Wallet

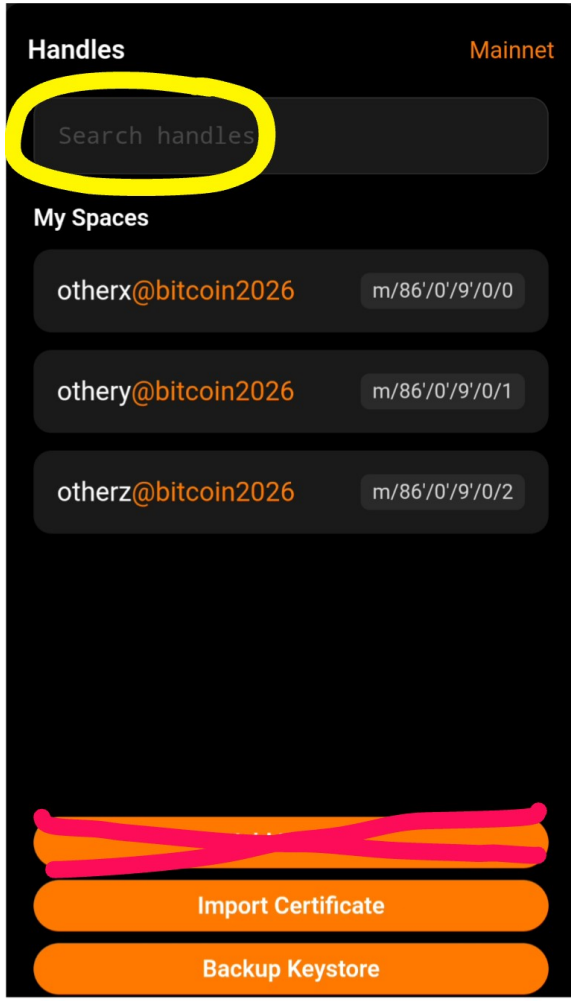


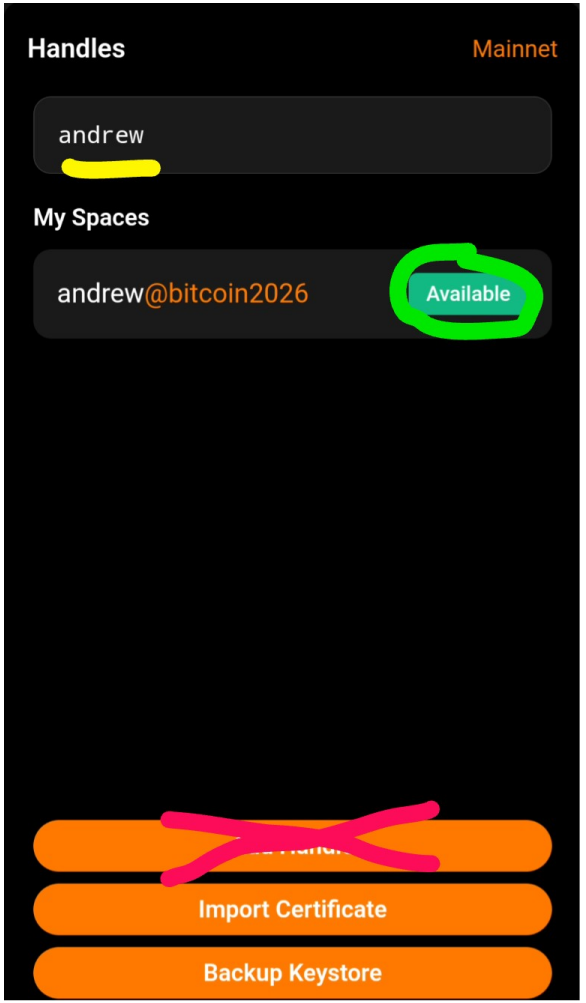
Bitcoin2026 Wallet

Nacho Bitcoin2026 Wallet

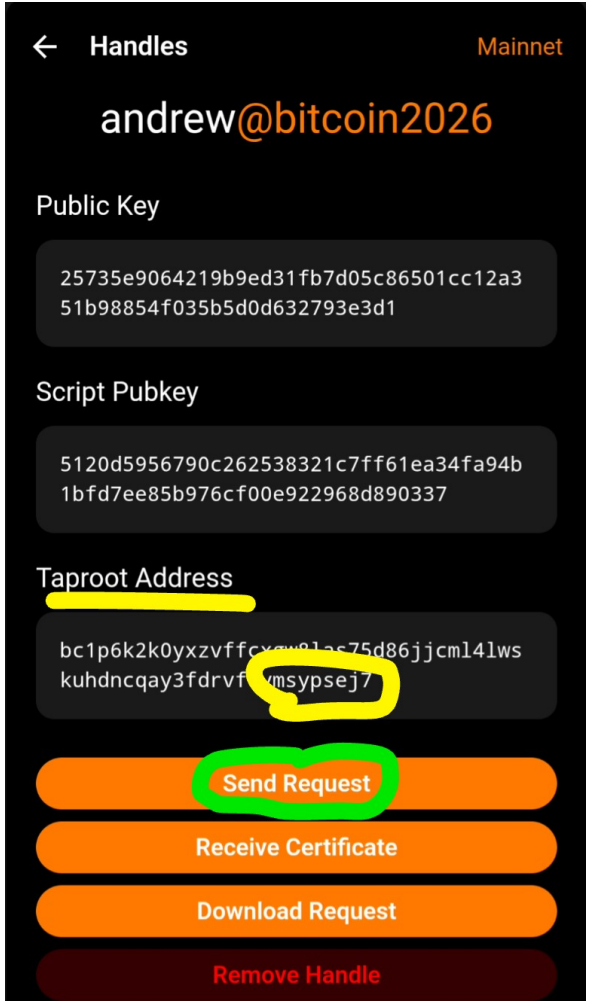


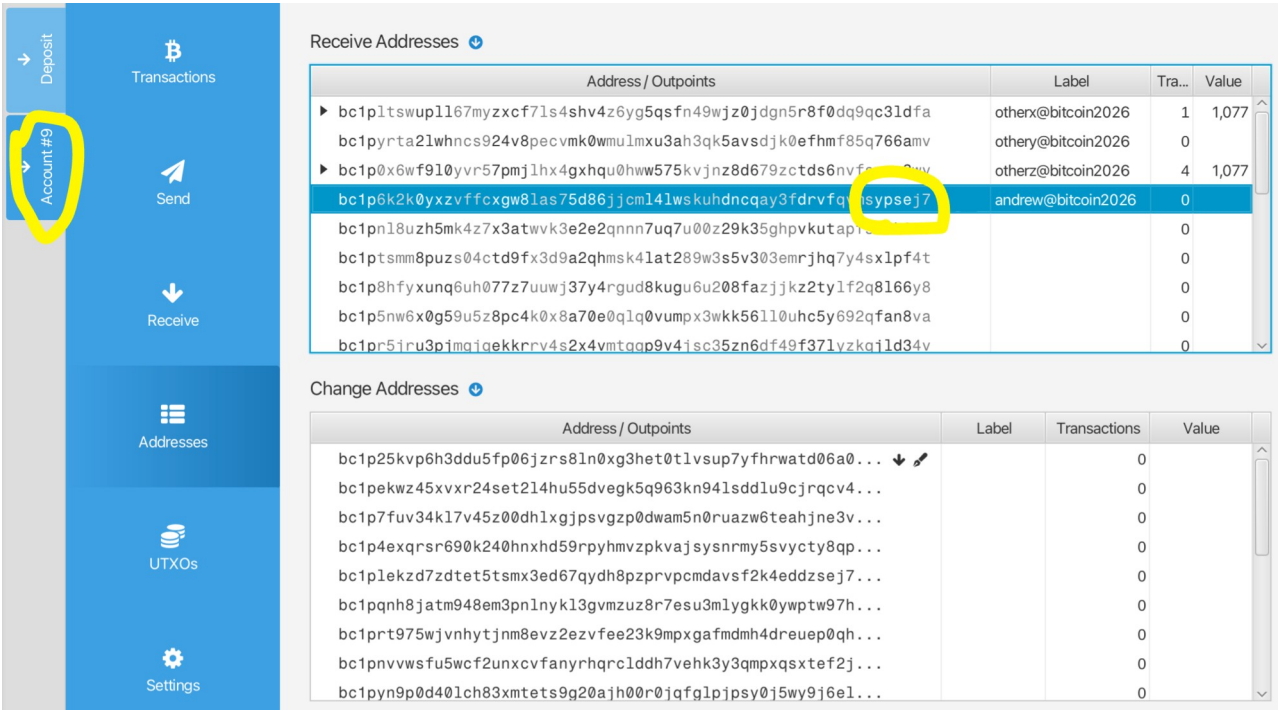
Workshop Exercise (1of4)





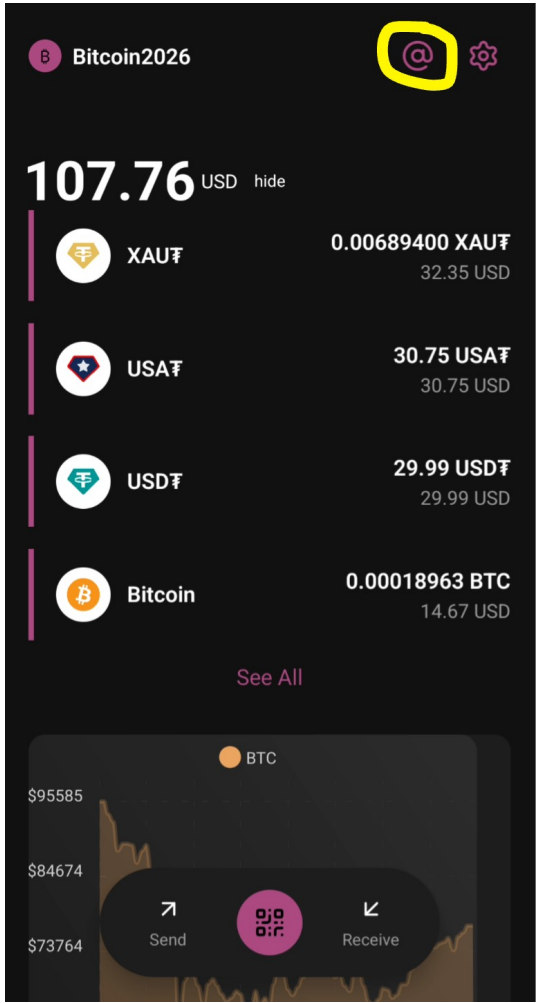
Workshop Exercise (2of4)

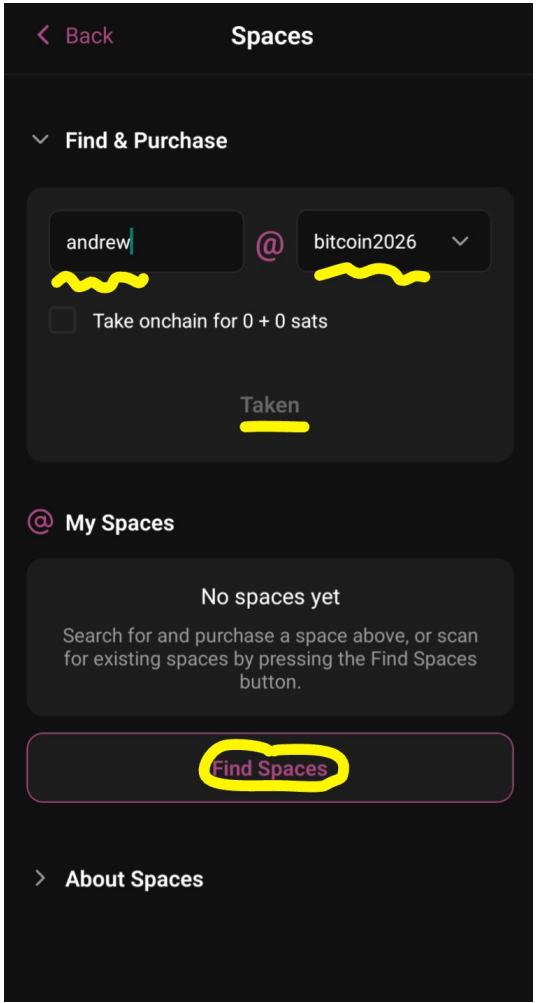


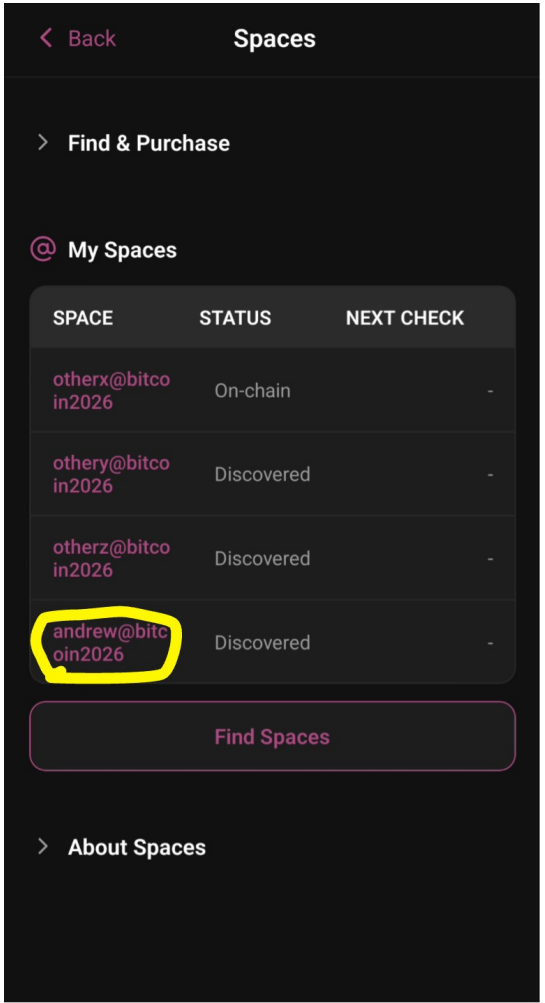


Intermission

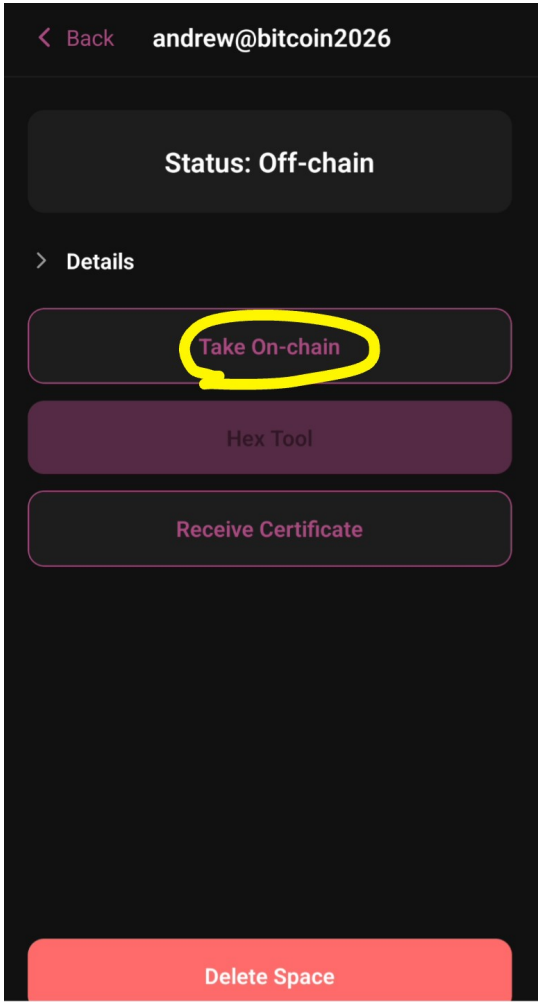
Spaces Wallet Demo (1of8)

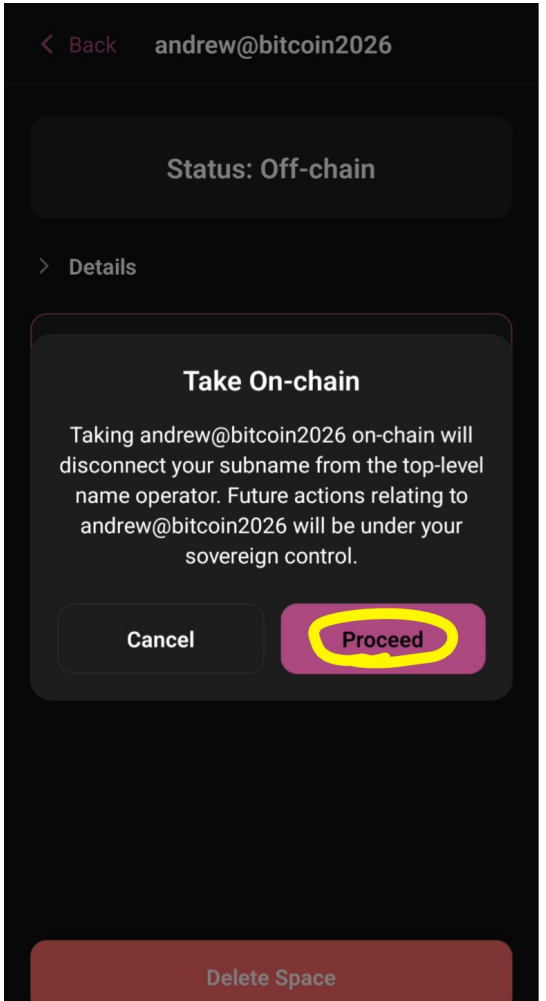


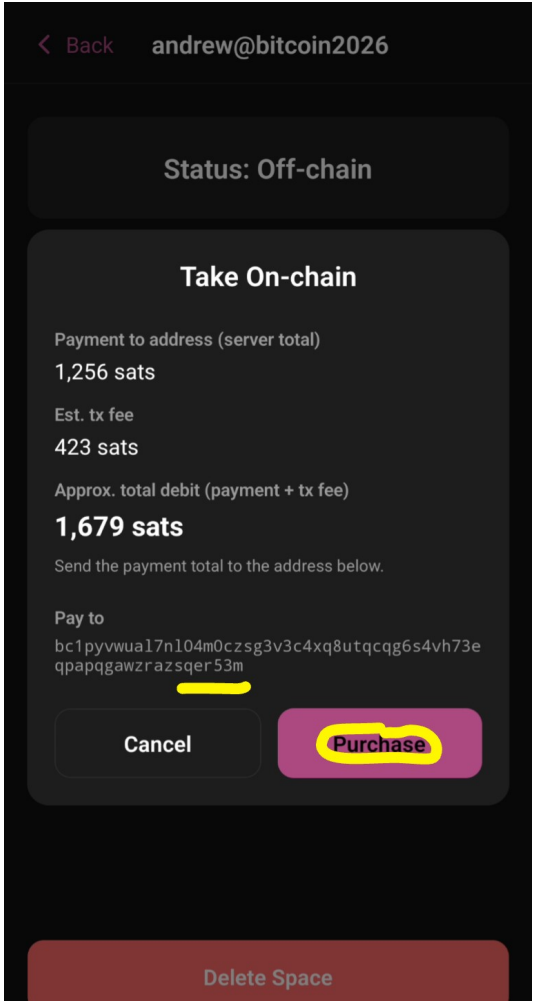




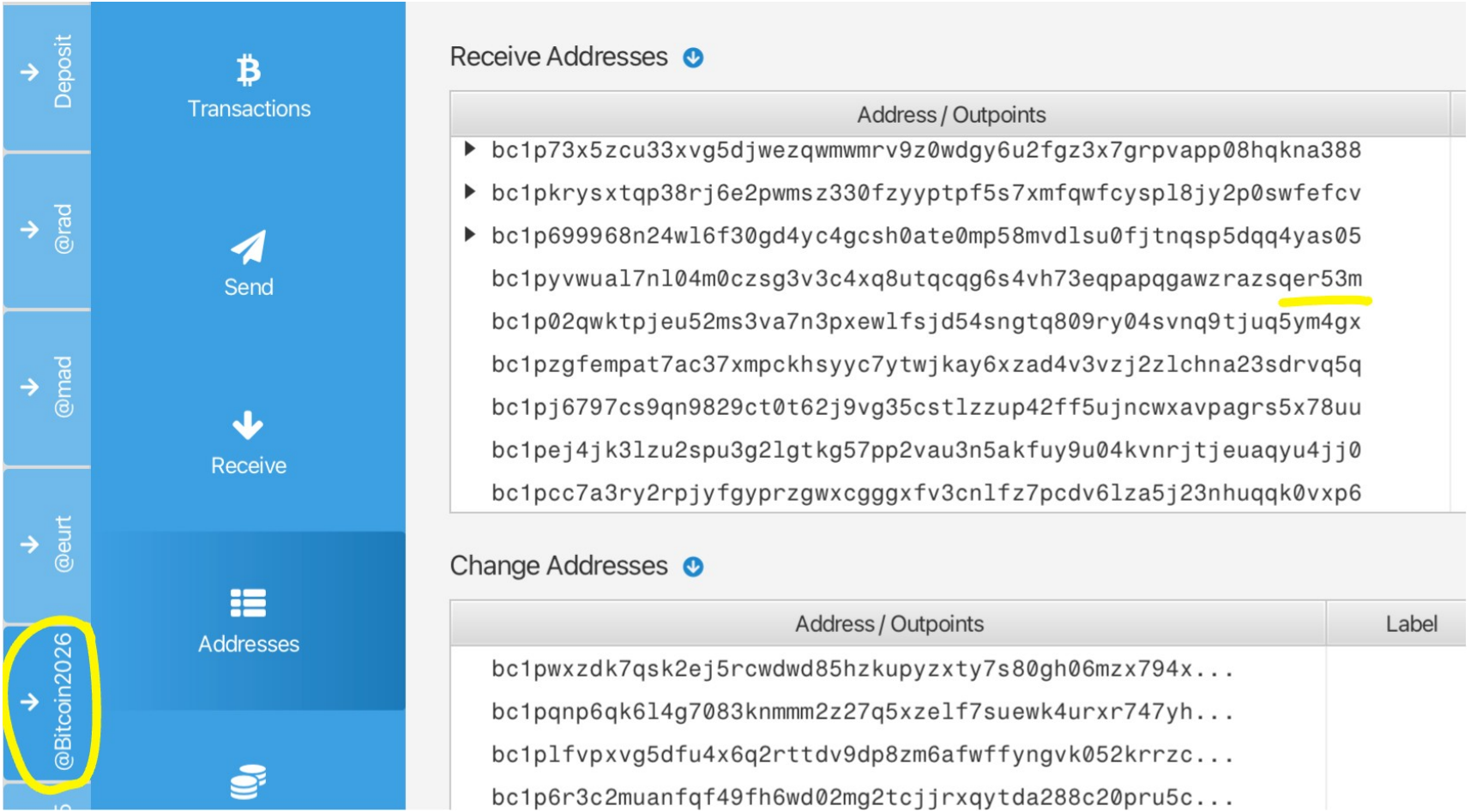
Spaces Wallet Demo (2of8)



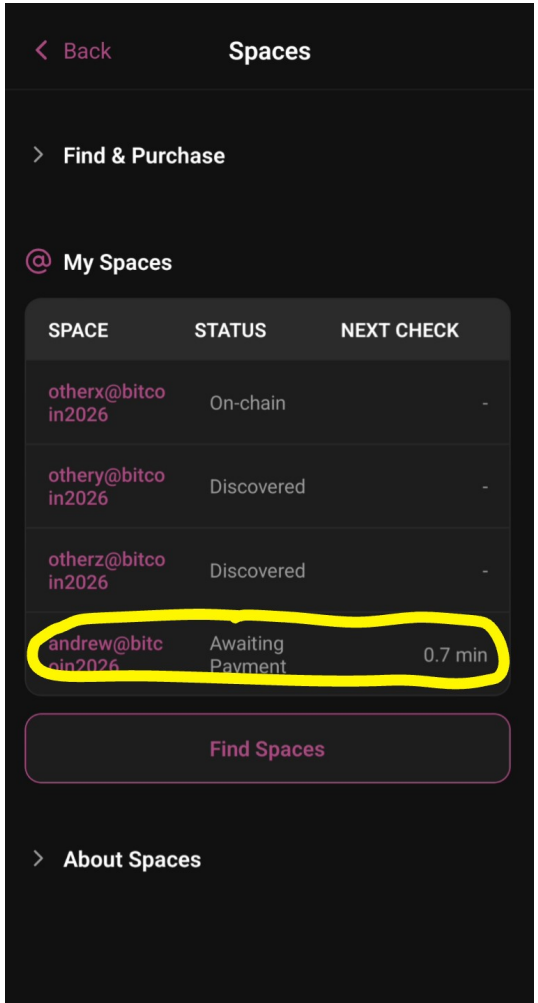


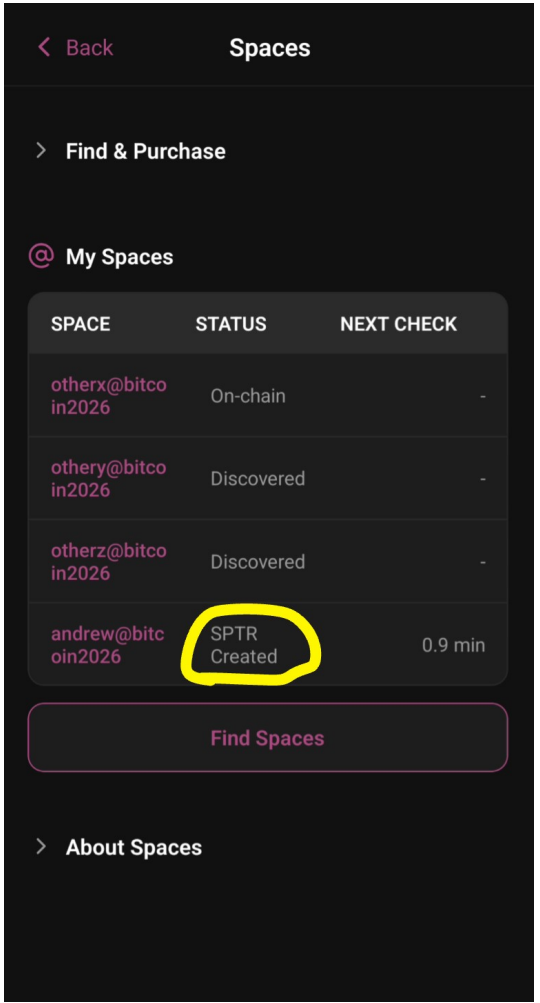


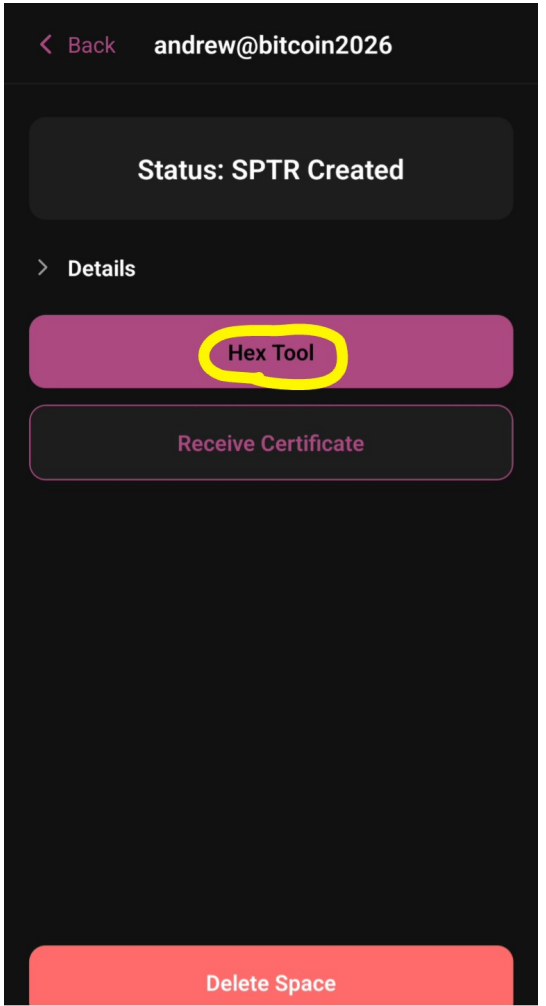
Spaces Wallet Demo (3of8)



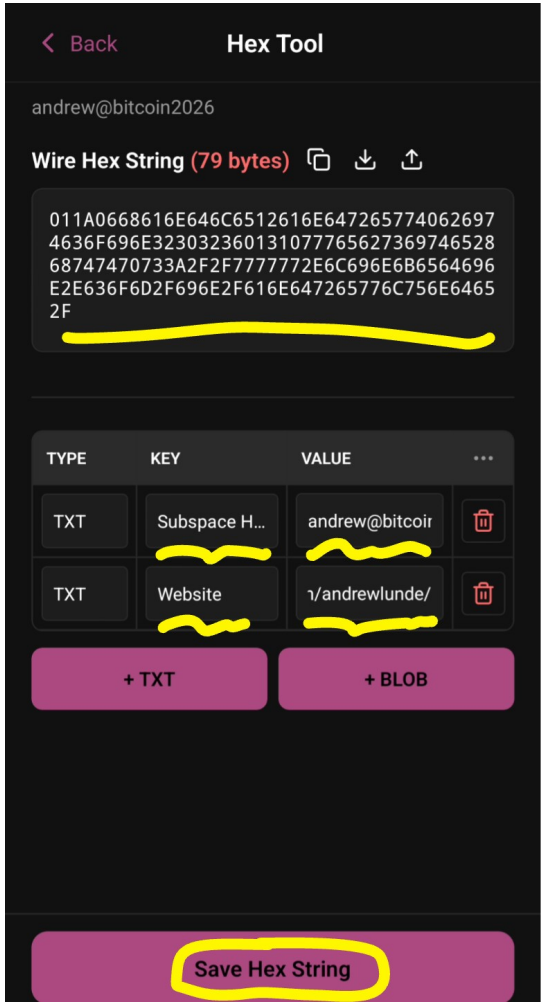
Spaces Wallet Demo (4of8)

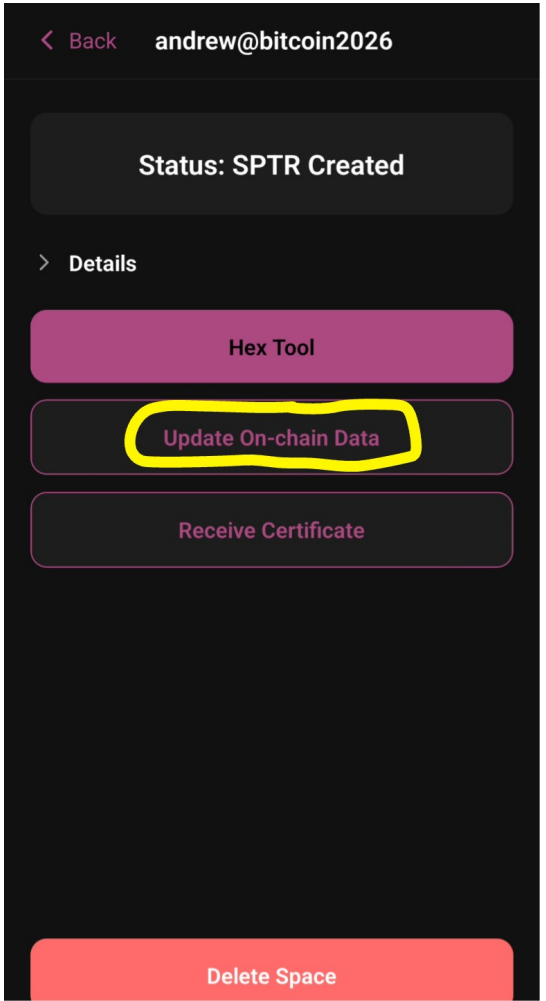


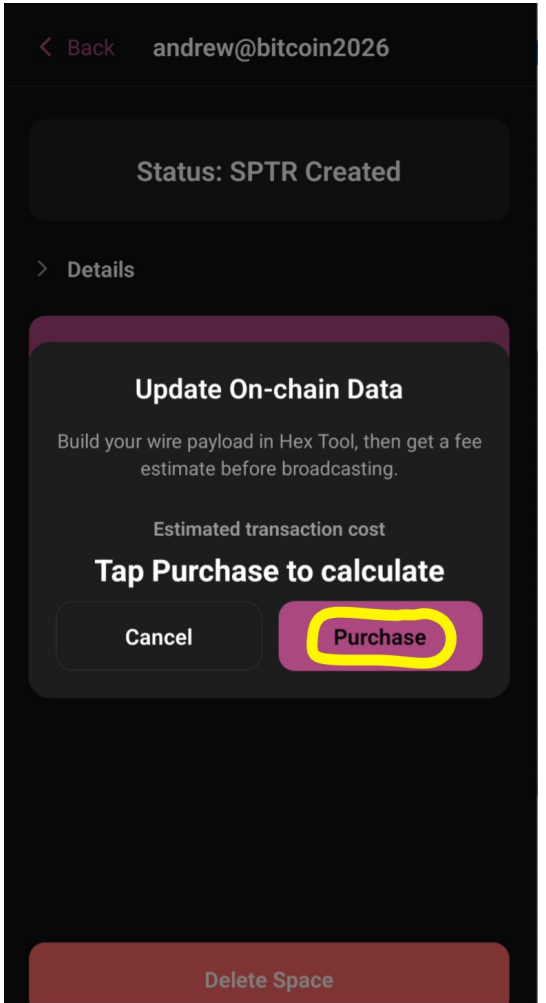




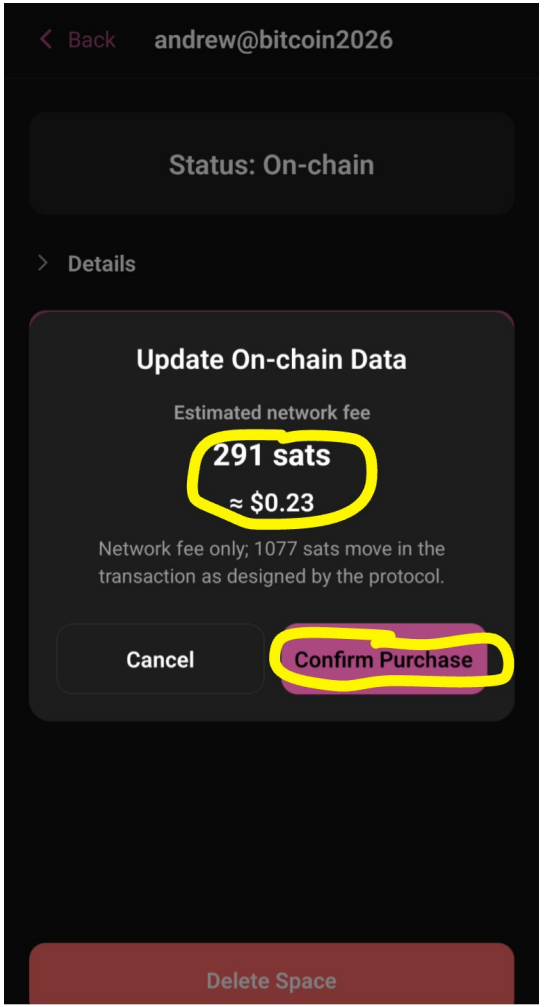
Spaces Wallet Demo (5of8)

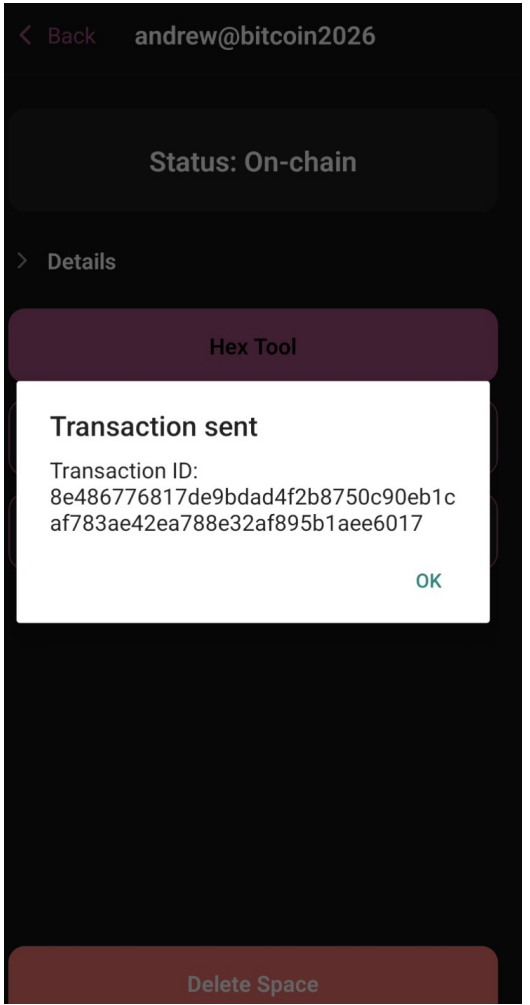






Spaces Wallet Demo (6of8)





Spaces Wallet Demo (7of8)

Deposit

Account #9

Transactions

Send

Receive Addresses

Address / Outpoints	Label	Trans...	Value
bc1pltswupll67myzxcf7ls4shv4z6yg5qsfn49wjz0jdgn5r8f0dq9qc3ldfa	otherx@bitcoin2026	1	1,077
bc1pyrta2lwhncs924v8pecvmk0wmulmxu3ah3qk5avsdjk0efhmf85q766amv	othery@bitcoin2026	0	
bc1p0x6wf9l0yvr57pmjlhx4gxhqu0hww575kvjnz8d679zctds6nvfsmav3wv	otherz@bitcoin2026	4	1,077
bc1p6k2k0yxzvffcxgw8las75d86jjcml4lwskuhdncqay3fdrvfqvmsypsej7	andrew@bitcoin2026	2	1,077
Received from output c56359a2..:0 on 17:17	andrew@bitcoin202...	1	~~1,077~~
Received from output 8e486776..:0 (Unconfirmed)	andrew@bitcoin202...	0	1,077
bc1pnl8uzh5mk4z7x3atwvk3e2e2qnnn7uq7u00z29k35ghpvkutapfs07k0r5		0	

- Tx [8e4867]
  - Inputs
    - c56359a2..:0
    - 2316fcde..:2
  - Outputs
    - bc1p6k2k0yxzvff...
    - Output #1
    - bc1pcp2p7nzg8k...

Output #1

Sends: 0 sats

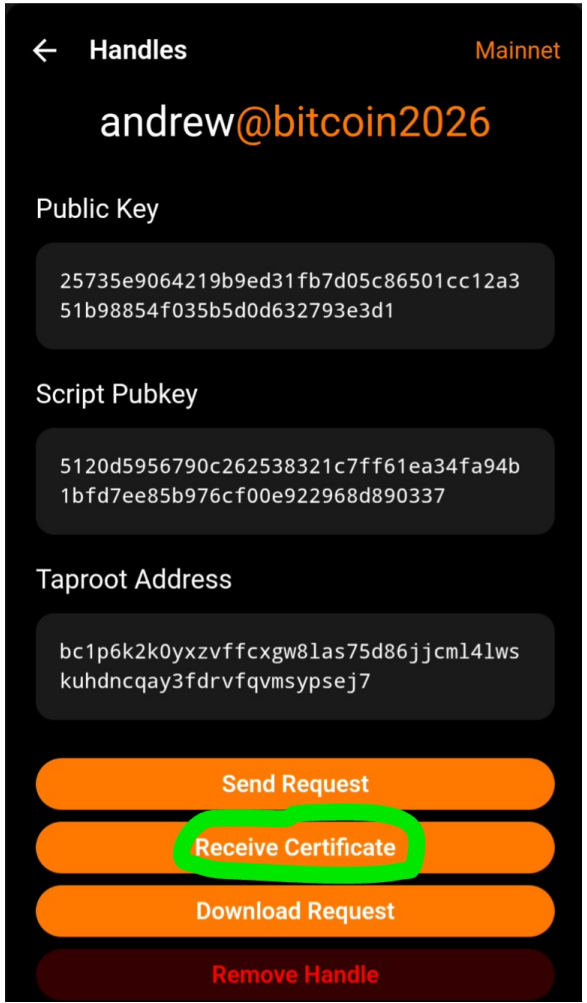
Spent? Unspent

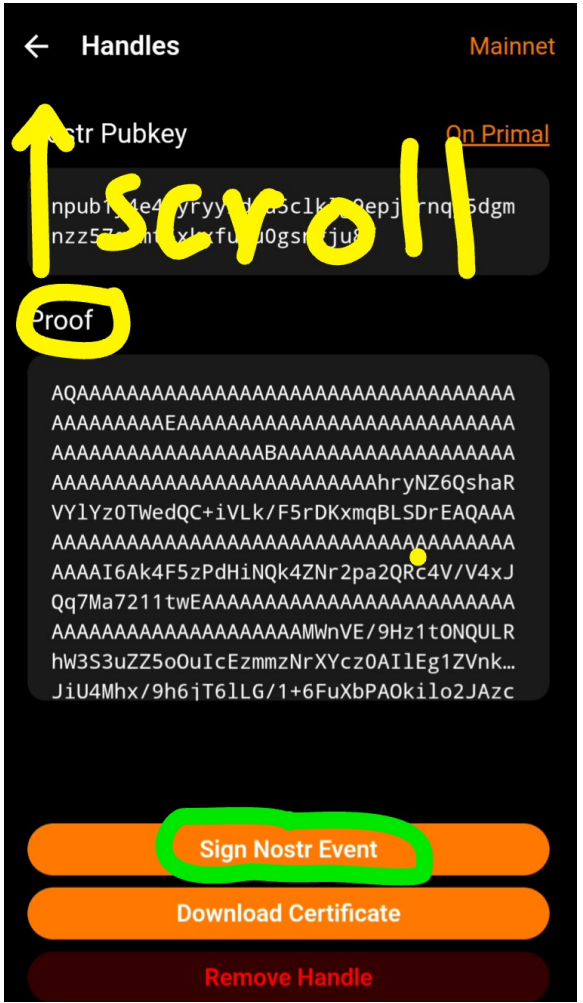
Script

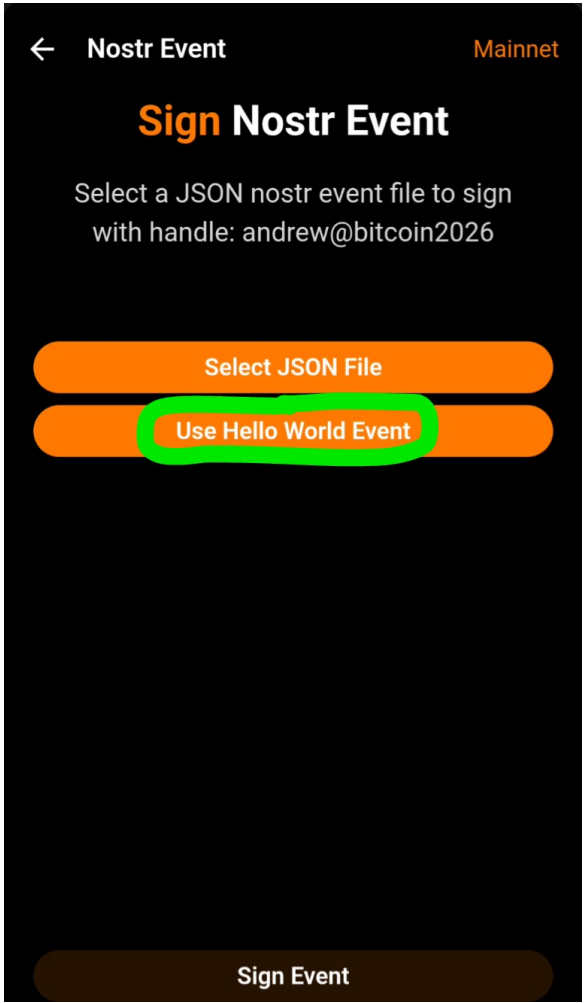
ScriptPubKey: `OP_RETURN OP_1 handle andrew@bitcoin2026 % website linkedin.com/in/andrewlunde/`

Resuming Workshop

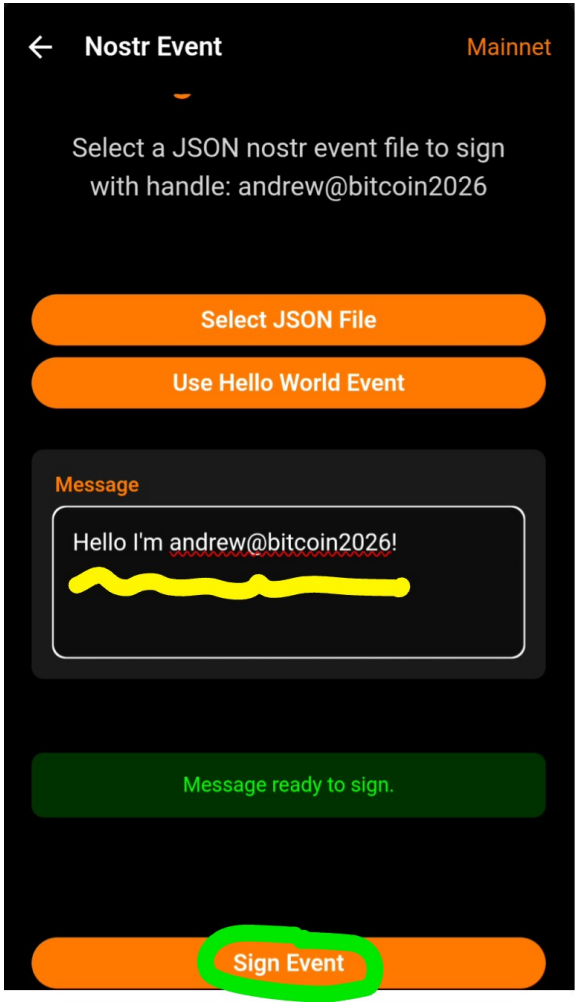
Workshop Exercise (3of4)

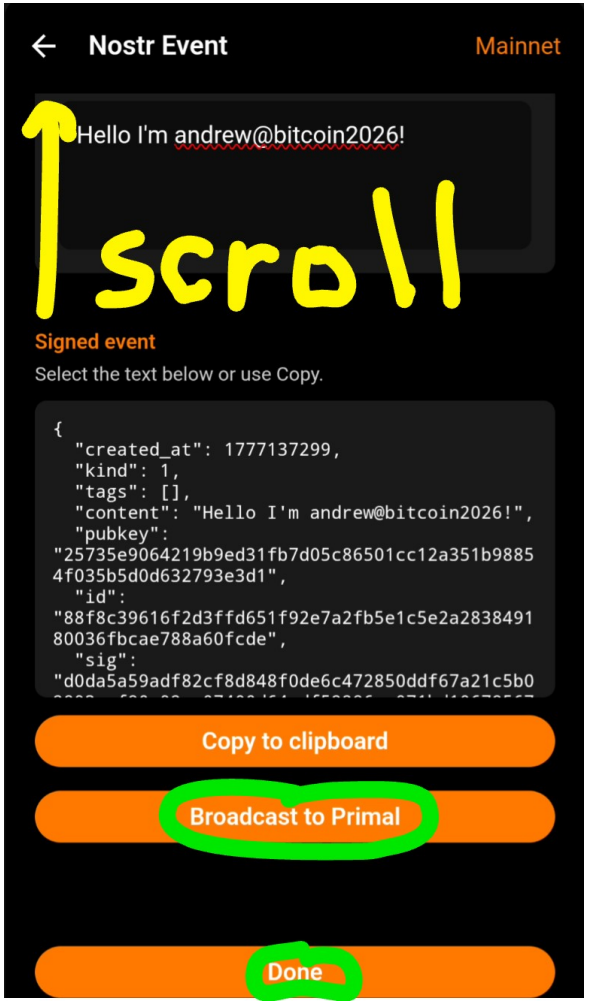


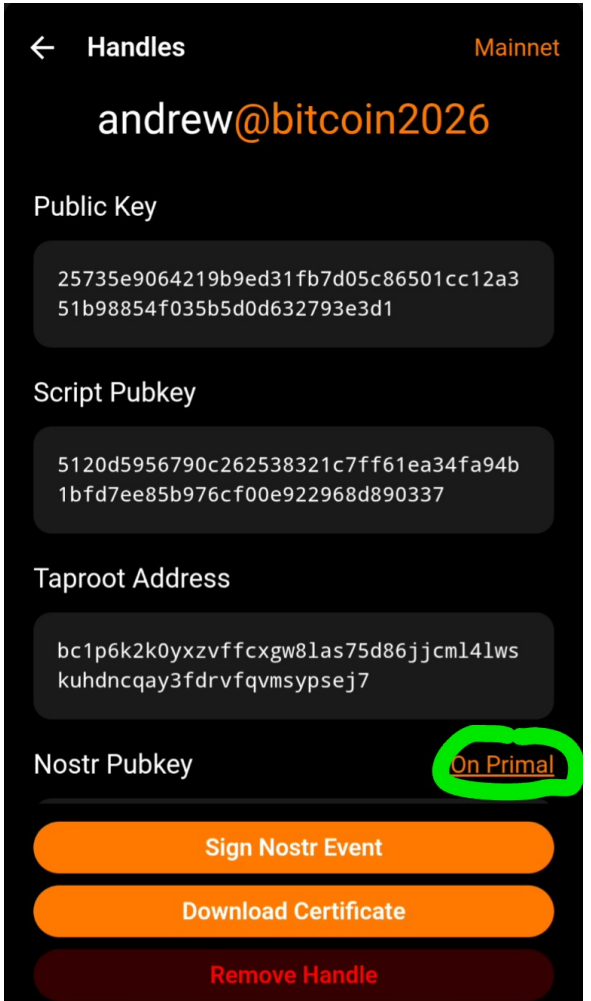




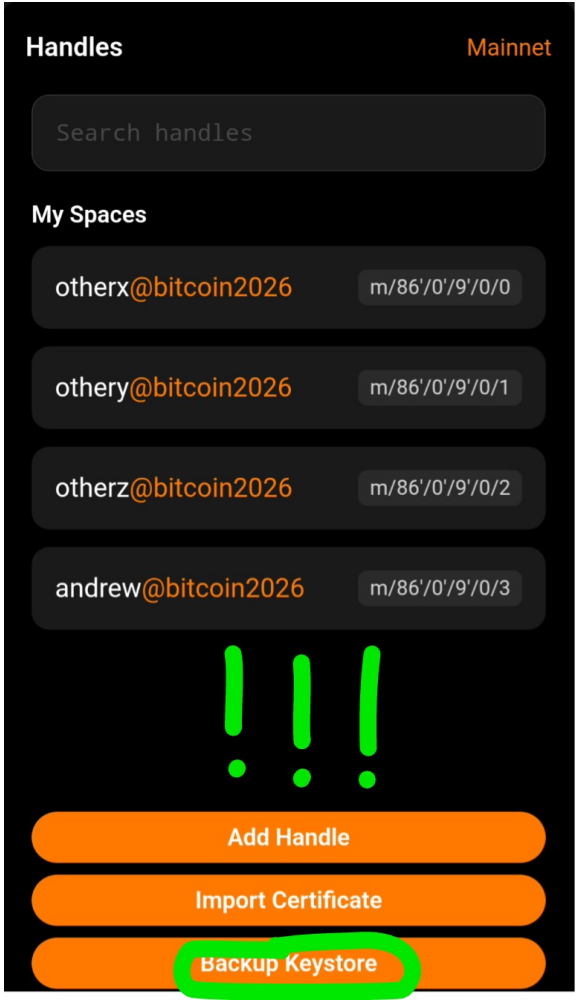
Workshop Exercise (4of4)







Please Backup your Keystore!



Sovereignty Attained

Spaces Wallet Demo (8of8)


```
"value": 1077,
"script_pubkey": "5120d5956790c262538321c7ff61ea34fa94b1bfd7ee85b97
"delegating_for": null,
"records": [
  {
    "type": "txt",
    "key": "handle",
    "value": [
      "andrew@bitcoin2026"
    ]
  },
  {
    "type": "txt",
    "key": "website",
    "value": [
      "linkedin.com/in/andrewlunde/"
    ]
  }
]
}
]
}
```

mempool .space

Explore the full Bitcoin ecosystem

Flow

Hide diagram

Inputs & Outputs

Details

bc1p6k2k0yxzvffcxgw8las75d86jjc... msypsej7	0.00001077 BTC	bc1p6k2k0yxzvffcxgw8las75d86... msypsej7	0.00001077 BTC
bc1pcp2p7nzg8kknr42w6yel8k7h... nq06nhac	0.00009352 BTC	OP_RETURN	0.00000000 BTC
		bc1pcp2p7nzg8kknr42w6yel8k7...nq06nhac	0.00009061 BTC
			0.00010138 BTC

Q & A

Backup Your Seed Phrase **AND** Keystore!